

Laboratory session 1

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25. September 2026

1 Laboratory session 1

The first experiment is using standing waves inside a long tube in order to measure the speed of sound inside the gas the tube is filled with. The tube and experiment settings are specified by the following data:

Tube length: $L = ? \pm (1.5mm?)$.

input signal: $V_0 = 35mV \pm ?$

Tubes:

K1: argon or CO_2 at $T = T_{room}$

K2: air at $T = T_{room}$

K3: air at $T = 70^\circ C = 243.15K$

K4: air at $T = 50^\circ C = 223.15$

$T_{room} = ?^\circ C = ?K$

We analyzed the properties of the gas in tube K?

2 Tasks

2.1 Measurement of ν_k

We measured the following resonance frequencies ν_k between $200Hz$ and $2kHz$:

K?: $\nu_k = [? \pm ?, \dots]$

The other groups measured:

K?: $\nu_k = [? \pm ?, \dots]$

K?: $\nu_k = [? \pm ?, \dots]$

K?: $\nu_k = [? \pm ?, \dots]$

2.2 How we estimated the error

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2.3 The best linear fit

In order to find the speed of sound in the gas, we want to use the equation $\nu_n = an + b$. However, we do not know which resonance frequency we are measuring, so we do not know which n belongs to which measured ν_n . This is why we set the reference of the measured ν_n to the first one measured. Therefore we get:

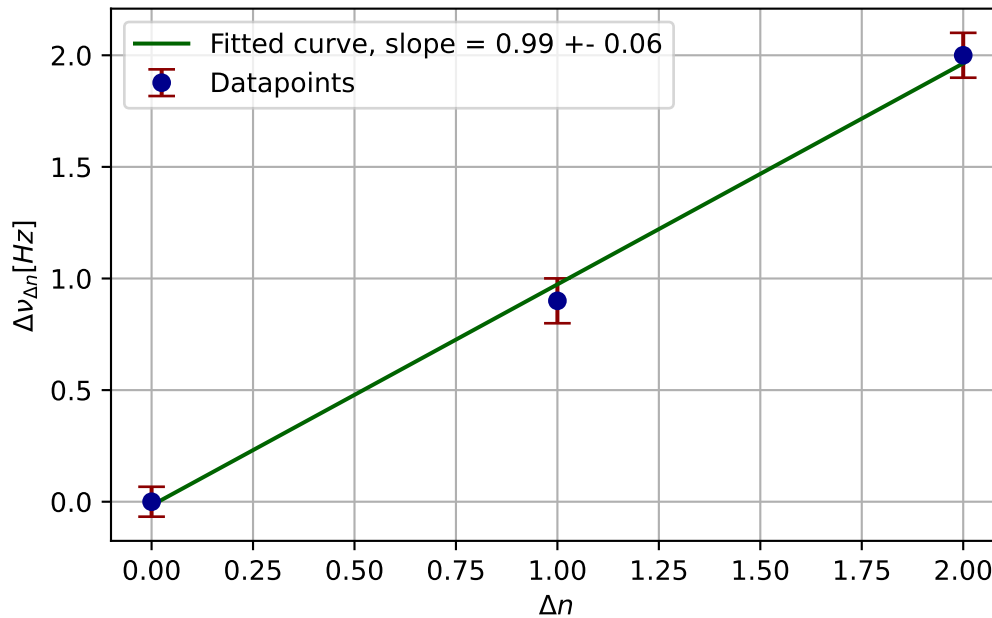
$$\nu_n - \nu_{n_0} = \Delta\nu_{\Delta n} = a\Delta n + b - (a(n_0) + b) = a(n - n_0) + b - b = a\Delta n = \frac{c}{2L}\Delta n$$

So the central relation is:

$$\Delta\nu_{\Delta n} = \frac{c}{2L}\Delta n$$

where Δn starts at 0 for the first measurement and for k measurements continues until $k - 1$ for the k^{th} resonance frequency. $\Delta\nu_{\Delta n}$ against Δn is plotted and evaluated in the following:

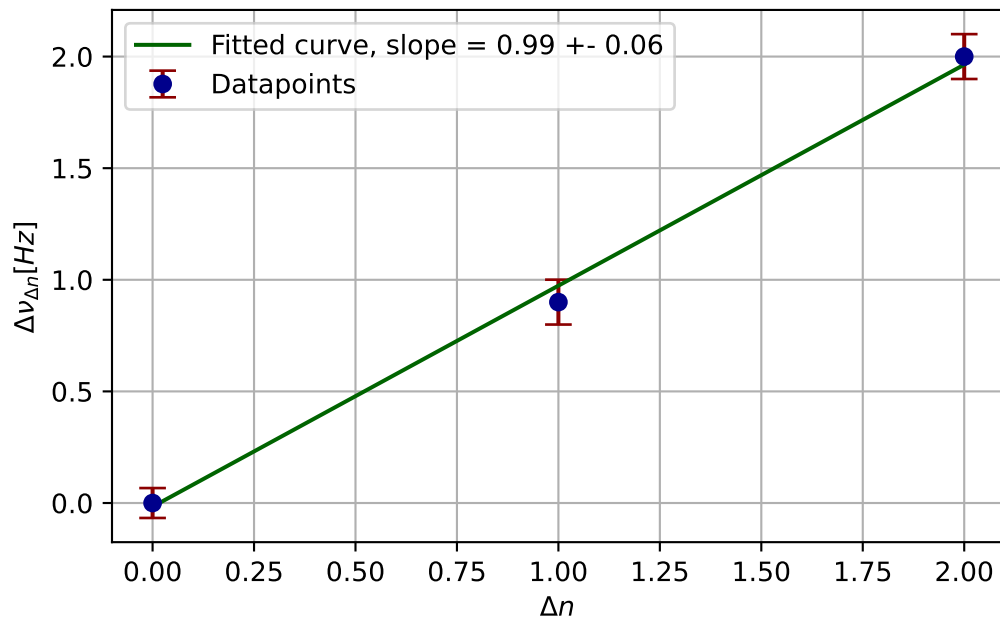
K?:



Given the slope of the fit curve, the velocity of sound can be calculated:

$$\text{speed of sound} = 1.98 \pm 0.23 \text{ m/s}$$

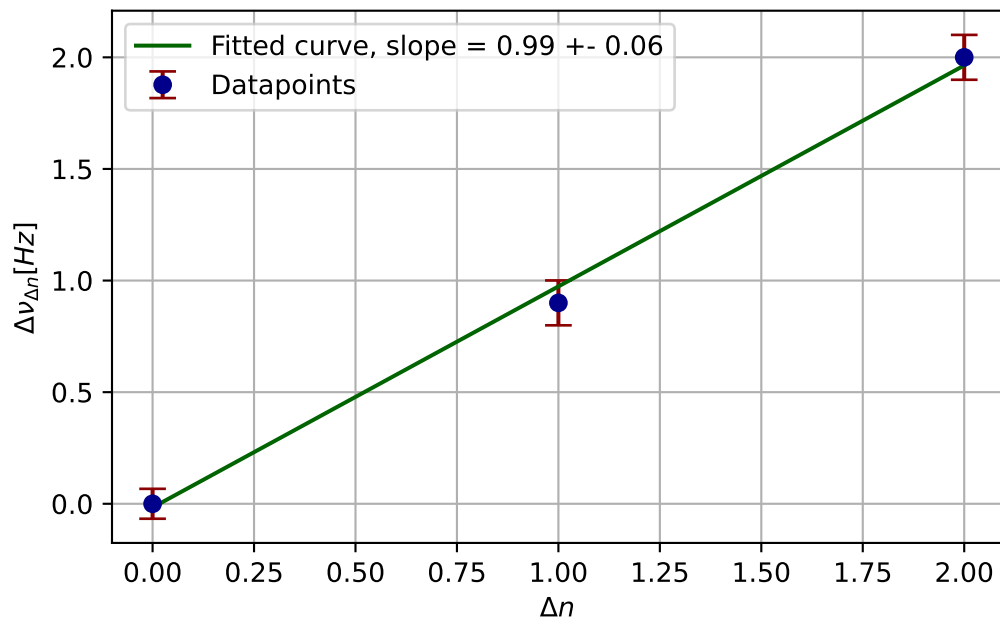
K?:



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 1.98 ± 0.23 m/s

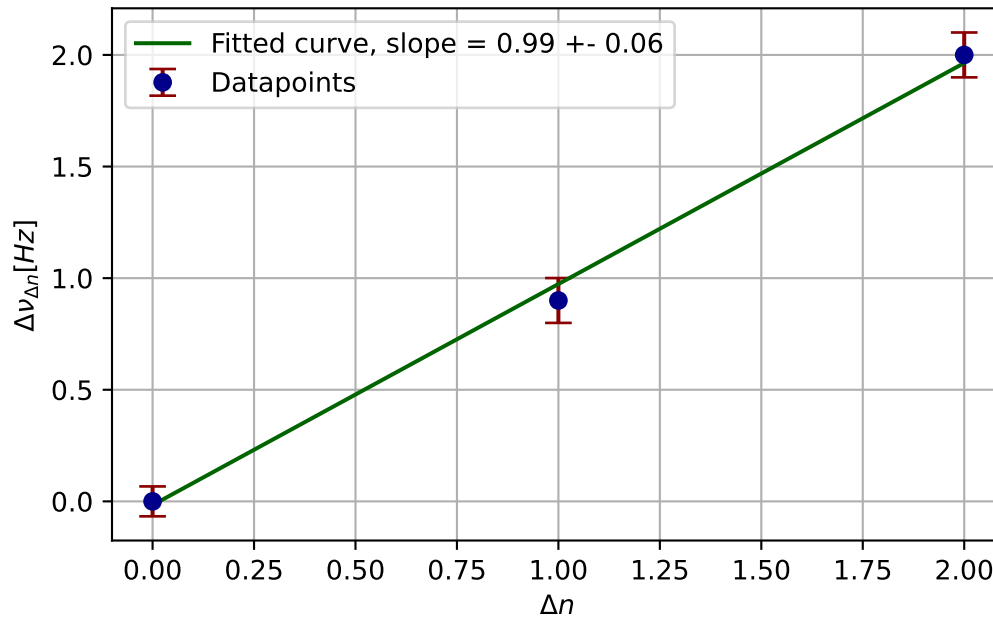
K?:



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 1.98 +- 0.23 m/s

K?:



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 1.98 +- 0.23 m/s

2.4 Most important contribution to uncertainty

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2.5 Degrees of freedom

For an ideal gas, the velocity of sound is given by

$$c(T) = \sqrt{\frac{(f + 2)RT}{fM_{mol}}}$$

where f are the degrees of freedom, R is the gas constant, T the temperature of the gas and M_{mol} the molar mass. This equation can be solved for f :

$$f = \frac{2RT}{c^2 M_{mol} + RT}$$

We calculate for f :

K?:

The calculated amount of degrees of freedom from the experiment is 1.94 ± 0.01

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In theory, the gases inside K?,... should have ? degrees of freedom. These consist of ...

2.6 Comparison to simulation and theory

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2.7 Phase diagram

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2.8 C_V , Z and degrees of freedom of a simulated lennard jones gas

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2.9 Different temperatures at fixed gas density

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2.10 Temperatures above the critical point

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2.11 Determination of C_V

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